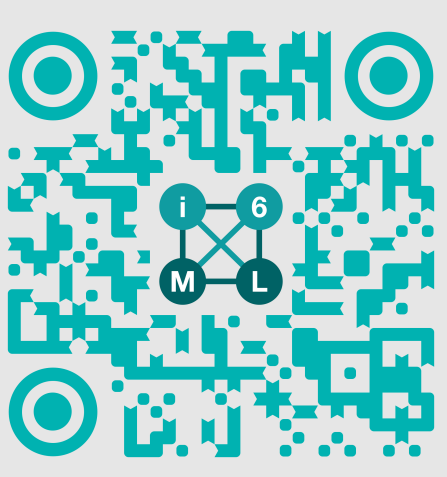
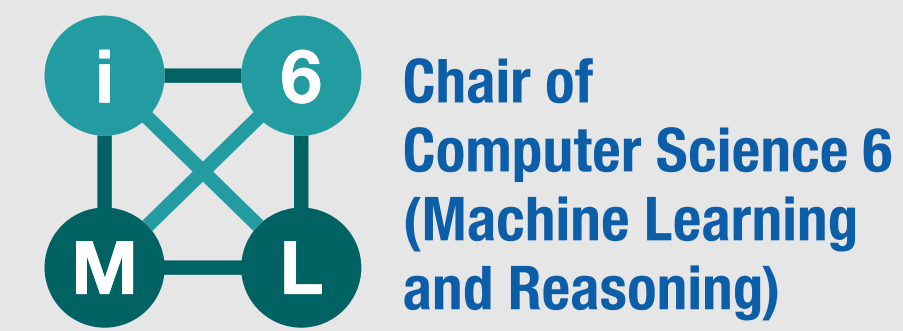




Learning Lifted Symbolic Planning Domains from Image Sequences

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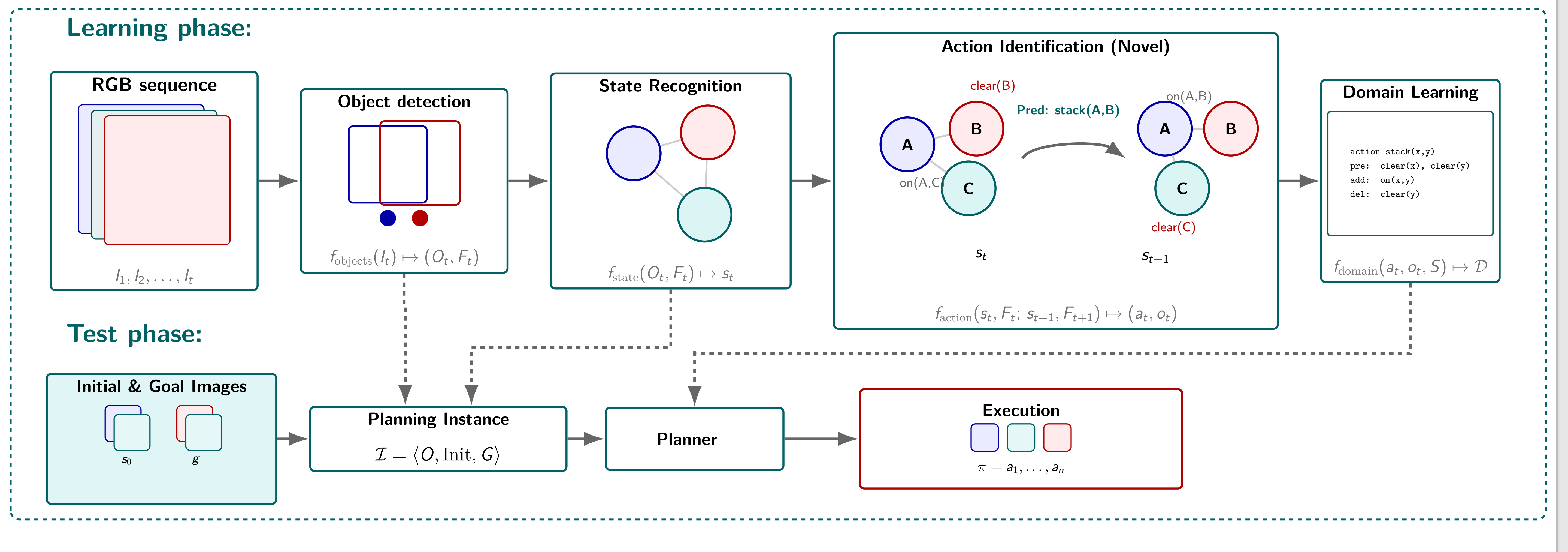
Problem

- Symbolic planners require action models, but these are usually **handcrafted** and expensive to build.
- Handcrafted symbolic models are **not naturally grounded** in raw *visual observations*.
- Existing visual model-learning approaches assume a **fixed number of objects**, *limiting generalization*.
- Many learning systems assume that training traces contain **fully grounded STRIPS actions** with *all arguments specified*.

Core Idea

- Detect objects** in images using an *object detector*.
- Represent each **visual scene as a scene graph** and infer unary and binary relations between detected objects using *dynamic scene graph detection transformer*.
- Identify actions** that caused the transition between the consecutive dynamic scene graphs with *dual-head GNNs*.
- Learn lifted action models** from the resulting state-action traces.
- Plan from images** by using the learned domain model with a standard symbolic planner *to solve larger instances*.

VISA: Visual-to-Symbolic Action Learning



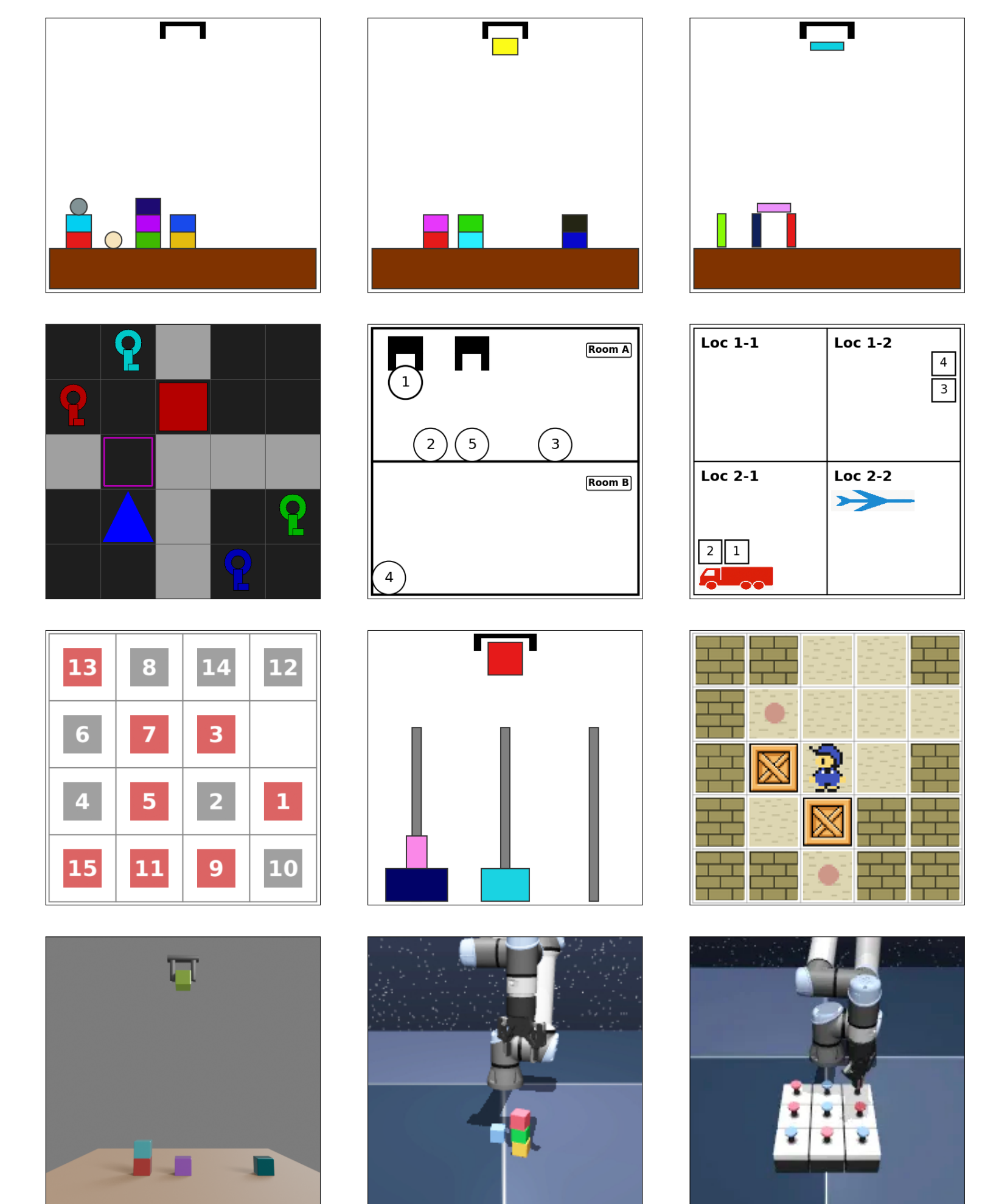
Main Results

Domain	Size			In-distribution					Generalization (OOD)					
	N_O	N_R	N_A	$[O_i, O_h]$	N_F	F1	ac_{args}	VAL	O_{train}	O_{test}	F1	ac_{type}	ac_{args}	VAL
BallHiding	4	8	6	5–12	8.6k	.99	99.8	1.00	5–8	10–11	.89	99.8	96.3	.80
Blocksworld	3	6	4	5–12	8.3k	1.00	99.4	1.00	4–8	9–11	1.00	99.6	100.0	1.00
BW-Realistic	3	6	4	5–12	8.3k	.99	93.9	1.00	4–8	9–11	.99	96.2	85.9	1.00
Dominoes	3	13	13	5–7	8.3k	1.00	100.0	1.00	N/A	N/A	N/A	N/A	N/A	N/A
Grid	1	16	20	25	5.2k	.98	99.8	1.00	5×5	6×6	.94	89.5	92.7	.85
Gripper	3	6	4	9–10	8.1k	1.00	100.0	1.00	7–8	9–10	1.00	100.0	100.0	1.00
Logistics	4	3	6	10–11	8.4k	1.00	91.2	1.00	8–10	11–12	1.00	99.2	95.5	1.00
N-Puzzle	3	5	4	18–32	8.8k	1.00	N/A	1.00	3×3	4×4	.84	97.4	N/A	.80
Sokoban-undir	1	4	2	25	6.8k	1.00	100.0	1.00	25	36	.83	79.5	38.1	SR
Sokoban-undir(#goals)	1	4	2	25	6.8k	1.00	100.0	1.00	25/2	25/3	.99	99.9	99.9	1.00
Hanoi	3	6	2	6–9	10.7k	.99	100.0	1.00	6–8	9	1.00	100.0	100.0	1.00

Performance results on synthetic domains.

high performance low performance

Visual Domains



Takeaways

- VISA** grounds visual **image sequences into relational symbolic states**, enabling *lifted domain learning and planning* from initial&goal images.
- Dynamic scene graphs allow **varying object counts**, but performance drops when *object detection fails* or perception is noisy.
- Scalability remains open** because relation extraction and *attention over object pairs* grow quickly with the number of objects.